

Practice Questions for Finals:

 Question Set 1 — Wheat Flour Output in a Milling Company

A milling company recorded its **annual wheat flour production (in metric tons)** from 2009 to 2023:

Year	Production (MT)
2009	2200
2010	2300
2011	2450
2012	2600
2013	2550
2014	2700
2015	2900
2016	3000
2017	3100
2018	3050
2019	3200
2020	3300
2021	3250
2022	3400
2023	3500

- Determine the trend using the **freehand method**.
- Calculate the **6-year central moving average**.
- Calculate the **7-year moving average**.
- Find the **semi-average trend value**.

Question Set 2 — Monthly Sales Volume of Packaged Juice (Company Data)

A beverage company recorded its **annual packaged juice sales (in '000 units)** from 2011 to 2025:

Year	Sales (000 units)
2011	800
2012	850
2013	900
2014	950
2015	920
2016	980
2017	1050
2018	1100
2019	1080
2020	1150
2021	1200
2022	1180
2023	1250
2024	1300
2025	1280

- Draw the **freehand trend line** for the data.
- Compute the **6-year central moving average**.
- Compute the **7-year moving average**.
- Determine the **semi-average trend**.

Question Set 3 — Annual Production of Cement Bags in a Manufacturing Plant

The following data show the **annual cement bag production (in million bags)** for a plant from 2008 to 2022:

Year	Production (million bags)
2008	4.5
2009	4.8
2010	5.0
2011	5.3
2012	5.1
2013	5.4
2014	5.6
2015	5.9
2016	6.0
2017	6.2
2018	6.4
2019	6.3
2020	6.6
2021	6.8
2022	7.0

- Estimate the trend using the **freehand method**.
- Calculate the **6-year central moving average**.
- Calculate the **7-year moving average**.
- Compute the **semi-average trend**.

✔ **Similar Question 1 — Customer Satisfaction Across Service Centers & Service Types**

A retail electronics company wants to evaluate whether customer satisfaction differs across its service centers under different types of service requests. The company collects **satisfaction scores (out of 10)** from five service centers across four service categories: **Repairs, Returns, Installations, and Technical Support**.

Service Center / Service Type	Repairs	Returns	Installations	Tech Support
Center A	8.2	7.1	8.9	8.4
Center B	7.8	6.5	8.3	7.9
Center C	8.5	7.4	9.1	8.7
Center D	7.9	6.9	8.5	8.0
Center E	8.3	7.0	8.8	8.5

The company wants to test:

- (a) Whether mean satisfaction scores differ by **service center**, assuming service type is homogeneous. (*Hint: One-way ANOVA*)
 - (b) Whether mean satisfaction scores differ by **service type**, assuming service center is homogeneous.
 - (c) Whether both **service center** and **service type** simultaneously affect satisfaction scores. (*Two-way ANOVA*)
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✔ **Similar Question 2 — Manufacturing Defect Rates Across Plants & Machine Settings**

A manufacturing company operates multiple plants and uses different machine calibration settings. The quality control department records the **defect rate (%)** from five plants across four machine settings: **Low Calibration, Medium Calibration, High Calibration, and Auto-Adjusted Mode**.

Plant / Machine Setting	Low	Medium	High	Auto
Plant A	3.2	2.4	1.8	2.1
Plant B	3.5	2.8	2.0	2.3
Plant C	3.1	2.3	1.7	2.0
Plant D	3.6	2.9	2.1	2.4
Plant E	3.3	2.5	1.9	2.2

The quality assurance analyst wants to test:

- (a) Whether average defect rates differ by **plant**, keeping machine setting homogeneous. (*One-way ANOVA*)
- (b) Whether average defect rates differ by **machine setting**, keeping plant homogeneous.
- (c) Whether **plant** and **machine setting** together significantly affect defect rates. (*Two-way ANOVA*)

 Similar Question 3 — Employee Productivity Across Departments & Work Modes

A consulting firm measures employee productivity (tasks completed per week) across five departments while employees work under four conditions: **On-site**, **Remote**, **Hybrid**, and **Flexible Hours**.

Department / Work Mode	On-site	Remote	Hybrid	Flexible
Dept A	52	48	55	50
Dept B	49	46	52	48
Dept C	54	50	57	52
Dept D	50	47	53	49
Dept E	53	49	56	51

The HR analytics team wants to test:

- (a) Whether mean productivity differs by **department**, assuming work mode is homogeneous. (*One-way ANOVA*)
- (b) Whether mean productivity differs by **work mode**, assuming department is homogeneous.

(c) Whether **department** and **work mode** interact to influence productivity. (*Two-way ANOVA*)

✔ **Question 1 — Training Satisfaction & Department**

An HR analyst wants to examine whether the **department in which employees work is associated with their level of satisfaction** with a new training program. Employees from three departments participated: **HR, IT, and Sales**. They rated the training as either **Satisfied** or **Not Satisfied**.

Department	Satisfied	Not Satisfied
HR	32	18
IT	45	25
Sales	38	29

Test whether training satisfaction is dependent on department.

✔ **Question 2 — Hybrid Work Preference & Age Group**

A workplace policy committee wants to check whether there is a **relationship between employee age group and preference for hybrid work arrangements**. Employees were categorized into three age groups: **Under 30, 30–45, and Over 45**. They indicated whether they **Prefer Hybrid Work** or **Prefer Full On-Site Work**.

Age Group	Prefer Hybrid	Prefer On-Site
Under 30	60	15
30–45	48	27
Over 45	25	35

Test whether hybrid work preference is related to age group.

✔ **Question 3 — Participation in Wellness Programs & Job Role**

A wellness coordinator wants to identify whether **employee job role is associated with participation** in the company's wellness initiatives. Employees from three job roles—**Administrative, Technical, and Managerial**—were surveyed regarding whether they **Participate** or **Do Not Participate**.

Job Role	Participate	Do Not Participate
Administrative	40	38
Technical	55	30
Managerial	28	32

Test whether participation in wellness programs is dependent on job role.

QUESTION 1 — High Positive Correlation (Direct Relationship)

Sales Revenue vs. Advertising Expenditure (Strong Direct Relation)

A retail chain wants to examine the **relationship between advertising expenditure and sales revenue** across 10 store locations:

Advertising (000\$)	10	15	18	20	25	28	30	35	38	40
Sales Revenue (000\$)	85	120	140	150	185	195	210	240	260	280

- I. Fit the regression model of sales on advertising and interpret the result.
- II. Estimate slope with a **99% CI**.
- III. Test the significance of slope.
- IV. Predict sales when advertising = **22 (000\$)**.
- V. Compute correlation coefficient and coefficient of determination and interpret.

QUESTION 2 — Moderate Positive Correlation (Direct Relationship)

Production Hours vs. Units Produced (Moderate Direct Relation)

A manufacturing company wants to study the **relationship between labor hours and total units produced** over 10 production batches:

Hours Worked	40	42	45	48	50	52	55	58	60	62
Units Produced	520	540	565	590	612	630	660	690	710	735

- I. Fit the regression model of output on labor hours and interpret.
 - II. Estimate **92% CI** for slope.
 - III. Test significance of slope.
 - IV. Predict units produced when labor hours = **47**.
 - V. Calculate correlation and R^2 and interpret.
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QUESTION 3 — Low Positive Correlation (Weak Direct Relationship)

Net Profit vs. Number of Employees (Weak Relation)

A firm wants to check whether **net profit is related to the number of employees** across 10 branches:

Employees	80	85	90	95	100	105	110	115	120	125
Net Profit (000\$)	210	215	225	220	230	235	240	238	245	250

- I. Fit the regression model of profit on employees and interpret.
 - II. Estimate **90% CI** for slope.
 - III. Test significance of slope.
 - IV. Predict net profit when employees = **102**.
 - V. Calculate correlation and R^2 and interpret .
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QUESTION 4 — Moderate Negative Correlation (Indirect Relationship)

Operating Costs vs. Net Income (Indirect Relation)

A company wants to examine whether **higher operating expenses are related to lower net income** across 10 months:

Operating Cost (000\$)	40	42	45	47	50	53	55	57	60	62
Net Income (000\$)	90	88	86	84	80	78	75	74	70	68

- I. Fit the regression model of net income on operating cost and interpret.
- II. Estimate **95% CI** for slope.
- III. Test the significance of the slope.
- IV. Predict net income when operating expenses = **48 (000\$)**.
- V. Compute correlation and R^2 and interpret .

✓ **QUESTION 1 — Accounting Department: Invoice Processing Times (Variance Comparison)**

A financial controller wants to evaluate whether the **consistency of invoice processing times** differs between two branches of the company. Faster **and more predictable** processing improves cash-flow management.

Two independent random samples of **10 processing times** (in hours) are collected:

Branch A — Central Office (hours)

4.8, 5.1, 4.6, 5.3, 4.9, 5.0, 5.2, 4.7, 5.4, 4.8

Branch B — Regional Office (hours)

5.6, 6.1, 5.9, 6.3, 5.7, 6.0, 6.2, 5.8, 6.4, 5.9

The CFO wants to know whether the **variability in processing time is different** between the two offices. Assume normality for both samples.

Task:

- 1) Test whether the variance of processing time in the Central Office differs from that in the Regional Office at the 5% significance level.
 - 2) Test whether the Average Processing time in Branch A is same then Branch B.
 - 3) Estimate the Variation of Time in Branch B using 99% confidence.
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✓ **QUESTION 2 — Manufacturing: Defect Rate Variability Across Two Production Lines**

A factory manager is comparing the **consistency of defect rates** between two production lines operated by different supervisors. Lower variability means more stable product quality.

Two independent random samples of defect percentages were taken over 14 days:

Line A — Supervisor 1 (% defects)

2.4, 2.6, 2.5, 2.7, 2.8, 2.4, 2.9, 2.6, 2.7, 2.5, 2.8, 2.6, 2.7, 2.5

Line B — Supervisor 2 (% defects)

3.1, 3.4, 3.2, 3.6, 3.5, 3.3, 3.7, 3.4, 3.6, 3.2, 3.5, 3.3, 3.4, 3.6

The quality control team wants to determine whether **Line B shows significantly higher variability** than Line A.

Assume the daily defect rates are normally distributed.

Task:

- 1) Conduct an appropriate statistical test to determine whether the variance of Line B differs from Line A.
 - 2) test that average defect rate in Line A is less than B.
 - 3) Estimate the variation of A using 95% confidence.
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 QUESTION 3 — Retail: Variability of Daily Sales Revenue Across Two Store Formats

A retail chain is analyzing whether the **variability of daily sales revenue** is different between its **mall stores** and **stand-alone stores**. Stable sales help with inventory and staffing planning.

Independent samples of **daily sales revenue (in \$1,000s)** over 12 days are recorded:

Mall Store Revenue (\$000)

52, 55, 54, 56, 53, 55, 57, 54, 56, 55, 53, 54

Stand-Alone Store Revenue (\$000)

48, 46, 49, 51, 50, 47, 52, 50, 48, 49, 51, 47

Management wants to know whether the **variability in revenue differs** between these store formats.

Assume normal distribution of daily sales.

Task:

Using a 5% significance level, test whether revenue variability differs between mall stores and stand-alone stores.

 Question 1 — Warehouse Order-Processing Time Variability

A logistics company states that the **variance in warehouse order-processing time should not exceed 9 minutes²** under standard operating procedures.

A supervisor collects a **random sample of 18 orders** and finds that the **sample variance is 13.2 minutes²**.

Task:

Using a 5% significance level, test whether the variance in order-processing time exceeds the company's acceptable limit.

 Question 2 — Payroll Department Error-Correction Time Variability

A payroll department claims that the **variance in error-correction time is equal to 2.5 minutes²**, based on historical performance.

A new audit reviews **a sample of 16 payroll corrections** and finds a **sample variance of 4.1 minutes²**.

Task:

At the 1% significance level, test whether the variance in correction time has changed from the claimed value.

Here's a version of your **four business situation questions with observations** included for each:

Question 1: Computer Hardware Industry Focus

Situation: A company in the computer hardware industry recently developed a new product. The team reports that **market testing** was the most difficult activity during product development.

Observation: Out of 56 sampled computer hardware companies, 48% reported market testing as the most difficult activity.

Question: Based on this observation, how does market testing compare to other development activities like completing the design or developing a business plan in terms of difficulty?

Question 2: Banking Industry Focus

Situation: A bank launched a new mobile app and reports that **market testing** was the most difficult part of the product development process.

Observation: Out of 89 sampled banks, 56% reported market testing as the most difficult activity.

Question: Construct a 99% confidence interval for the proportion of banks that consider market testing the most difficult activity. Interpret the results in a business context.

Question 3: Industry Comparison

Situation: A researcher wants to compare the computer hardware and banking industries regarding the difficulty of market testing in product development.

Observation: 48% of 56 computer hardware companies and 56% of 89 banks reported market testing as the most difficult activity.

Question: Using a significance level of 0.02, test whether there is a significant difference in the responses between the two industries.

Question 4: Strategic Decision-Making

Situation: A business analyst is advising firms on resource allocation during product development. They notice differences in market testing difficulty between industries.

Observation: Banks report market testing as more difficult (56%) compared to computer hardware companies (48%).

Question: Based on this observation, should the analyst conclude there is a statistically significant difference between the two industries? How could this insight guide the allocation of resources for product development?